

Sid Mathur's Thematic ETF Screener – Daily Report

2026-05-12

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Action Center**Run date:** 2026-05-12 **Universe:** 668 ETFs / 26 clusters**Full PC1:** LONG since 2026-04-13 (D+20, 4.0%) **Clusters long:** 14/26 **ETFs long:** 448/668 (67%)**Today's events****BUY (21)**

level	id	name	category	cluster_signal	spread_buy	prob_buy	nss
etf	SIVR			FLAT	2.057	0.95	0.796
etf	SLV			FLAT	1.939	0.95	0.794
etf	KWEB			FLAT	0.209	0.94	0.656
etf	LTPZ			LONG	0.047	0.87	0.292
etf	PPLT			FLAT	0.746	0.66	0.155
etf	SILJ			FLAT	0.041	0.66	0.141
etf	BAB			LONG	0.004	0.74	0.098
etf	SIL			FLAT	0.735	0.68	0.096
etf	SUB			FLAT	0.019	0.90	0.003
etf	VMBS			LONG	0.055	0.78	-0.083
etf	BOND			LONG	0.090	0.81	-0.105
etf	MBB			LONG	0.078	0.70	-0.169
cluster	cluster_5	Mixed: BND, AGG, SCHZ	Mixed		0.035	0.70	-0.177
etf	GVI			LONG	0.034	0.70	-0.224
etf	AGG			LONG	0.028	0.65	-0.232
etf	BND			LONG	0.017	0.64	-0.256
etf	SPAB			LONG	0.003	0.59	-0.299
etf	SCHZ			LONG	0.002	0.58	-0.343
etf	NANR			FLAT	0.328	0.35	-0.351
etf	SPMB			LONG	0.005	0.53	-0.358
etf	AGZ			LONG	0.013	0.45	-0.526

SELL (4)

level	id	name	category	cluster_signal	spread_sell	prob_sell	nss
etf	LVHD			LONG	-0.260	0.99	-0.952
etf	FDL			LONG	-0.277	0.91	-0.883
etf	PKW			LONG	-0.691	0.97	0.033
etf	IPAY			FLAT	-0.544	0.82	0.165

Imminent (cluster level, $p > 50\%$)**Imminent BUY (top 5)**

id	name	category	spread_buy	prob_buy	nss
cluster_23	Mixed: GSY, FTSM, SHV	Mixed	0.393	1	1.000
cluster_22	Mixed: SRLN, BKLN, FTSL	Mixed	1.305	1	1.000
cluster_3	Mixed: ONEQ, IGM, ILCG	Mixed	17.497	1	1.000
cluster_8	Mixed: IEMG, EEM, SPEM	Mixed	11.512	1	1.000
cluster_4	Mixed: IWB, ITOT, SCHB	Mixed	8.861	1	0.987

Imminent SELL (top 5)

id	name	category	spread_sell	prob_sell	nss
cluster_26	Mixed: UUP, SQQQ	Mixed	-1.324	1.00	-1.000
cluster_17	Mixed: VDC, FSTA, XLP	Mixed	-0.034	0.74	-0.232
cluster_15	Mixed: DBP, GLTR, GDXJ	Mixed	0.401	0.61	-0.193
cluster_11	Mixed: VHT, FHLC, FXH	Mixed	-0.362	0.87	0.021
cluster_18	Mixed: ILF, EWZ, ECH	Mixed	-0.544	0.91	0.072

Full Sample PC1



Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages (last 12 months).

Current state: LONG since 2026-04-13 (20 days, 4.0%)

5dma: 140.590 20dma: 138.541 50dma: 134.180

Spread vs 50dma: +6.410 Spread vs 20dma: +2.050

P(buy crossover within 5 days): 100.0% P(sell): 4.3%

Top 10 PC1 weights

	ticker	weight	name	category
102	ACWI	0.0021		
113	AOA	0.0021		
133	SPGM	0.0021		
137	VT	0.0021		
46	SPY	0.0020		
116	AOR	0.0020		
120	URTH	0.0020		

122	CRBN	0.0020
180	RSP	0.0020
264	IMCG	0.0020

Position Sheet

Replicate each currently-long cluster via its 3-5 member basket. OLS weights sum to ~1 per cluster for unit-variance replication; scale each cluster's basket to your target cluster allocation. sticky=yes means members are unchanged from the prior run (only weights drift); sticky=no means a fresh greedy selection.

Active positions (long clusters)

ticker	cluster	theme	cluster_nss	days_long	cum_return	basket_weight	basket_cor	sticky	etf_signal
IEMG	cluster_8	Mixed: IEMG, EEM, SPEM	1.000	21	9.9%	0.754	1.00	no	LONG
SRLN	cluster_22	Mixed: SRLN, BKLN, FTSL	1.000	24	1.2%	0.346	0.97	no	LONG
FLOT	cluster_22	Mixed: SRLN, BKLN, FTSL	1.000	24	1.2%	0.322	0.97	no	FLAT
LQDH	cluster_22	Mixed: SRLN, BKLN, FTSL	1.000	24	1.2%	0.319	0.97	no	LONG
DEM	cluster_8	Mixed: IEMG, EEM, SPEM	1.000	21	9.9%	0.168	1.00	no	LONG
FEMS	cluster_8	Mixed: IEMG, EEM, SPEM	1.000	21	9.9%	0.067	1.00	no	FLAT
IWB	cluster_4	Mixed: IWB, ITOT, SCHB	0.987	20	5.9%	0.636	1.00	no	LONG
VT	cluster_4	Mixed: IWB, ITOT, SCHB	0.987	20	5.9%	0.233	1.00	no	LONG
IYC	cluster_4	Mixed: IWB, ITOT, SCHB	0.987	20	5.9%	0.086	1.00	no	LONG
IWD	cluster_6	Mixed: IWD, SCHV, VONV	0.919	20	2.6%	0.459	1.00	no	LONG
CFA	cluster_6	Mixed: IWD, SCHV, VONV	0.919	20	2.6%	0.310	1.00	no	LONG
DGRO	cluster_6	Mixed: IWD, SCHV, VONV	0.919	20	2.6%	0.222	1.00	no	LONG
GSG	cluster_21	Mixed: GSG, DBC, PDBC	0.918	84	47.8%	0.478	1.00	no	FLAT
PDBC	cluster_21	Mixed: GSG, DBC, PDBC	0.918	84	47.8%	0.353	1.00	no	LONG
DBO	cluster_21	Mixed: GSG, DBC, PDBC	0.918	84	47.8%	0.179	1.00	no	FLAT
VB	cluster_2	Mixed: VB, SCHM, IJH	0.849	20	3.7%	0.687	1.00	no	LONG
PRN	cluster_2	Mixed: VB, SCHM, IJH	0.849	20	3.7%	0.158	1.00	no	LONG
RSPD	cluster_2	Mixed: VB, SCHM, IJH	0.849	20	3.7%	0.122	1.00	no	FLAT
SCHH	cluster_14	Mixed: SCHH, VNQ, FREL	0.816	21	4.2%	0.541	1.00	no	LONG
RWR	cluster_14	Mixed: SCHH, VNQ, FREL	0.816	21	4.2%	0.336	1.00	no	LONG
INDS	cluster_14	Mixed: SCHH, VNQ, FREL	0.816	21	4.2%	0.100	1.00	no	LONG
TIP	cluster_19	Mixed: TDTF, TIPX, TIP	0.787	22	0.8%	0.627	1.00	no	LONG
DVY	cluster_10	Mixed: DVY, SPYD, SDOG	0.787	14	0.6%	0.571	0.99	no	LONG
FDL	cluster_10	Mixed: DVY, SPYD, SDOG	0.787	14	0.6%	0.263	0.99	no	FLAT
STPZ	cluster_19	Mixed: TDTF, TIPX, TIP	0.787	22	0.8%	0.194	1.00	no	LONG
SPIP	cluster_19	Mixed: TDTF, TIPX, TIP	0.787	22	0.8%	0.136	1.00	no	LONG
AML	cluster_10	Mixed: DVY, SPYD, SDOG	0.787	14	0.6%	0.129	0.99	no	LONG
JNK	cluster_9	Mixed: JNK, HYG, HYL	0.489	20	0.7%	0.487	0.99	no	LONG
PFF	cluster_9	Mixed: JNK, HYG, HYL	0.489	20	0.7%	0.257	0.99	no	LONG
PCY	cluster_9	Mixed: JNK, HYG, HYL	0.489	20	0.7%	0.115	0.99	no	FLAT
VBR	cluster_7	Mixed: VBR, MDYV, IVOV	0.380	21	2.9%	0.557	1.00	no	LONG
SPSM	cluster_7	Mixed: VBR, MDYV, IVOV	0.380	21	2.9%	0.222	1.00	no	LONG
KBE	cluster_7	Mixed: VBR, MDYV, IVOV	0.380	21	2.9%	0.164	1.00	no	LONG
VHT	cluster_11	Mixed: VHT, FHL, FXH	0.021	3	-2.0%	0.472	0.99	no	FLAT

FXH	cluster_11	Mixed: VHT, FHLC, FXH	0.021	3	-2.0%	0.363	0.99	no	FLAT
XBI	cluster_11	Mixed: VHT, FHLC, FXH	0.021	3	-2.0%	0.185	0.99	no	FLAT
BND	cluster_5	Mixed: BND, AGG, SCHZ	-0.177	0	0.0%	0.505	1.00	no	LONG
BIV	cluster_5	Mixed: BND, AGG, SCHZ	-0.177	0	0.0%	0.219	1.00	no	FLAT
BLV	cluster_5	Mixed: BND, AGG, SCHZ	-0.177	0	0.0%	0.134	1.00	no	FLAT
VDC	cluster_17	Mixed: VDC, FSTA, XLP	-0.232	6	-2.3%	0.527	1.00	no	LONG
FUTY	cluster_17	Mixed: VDC, FSTA, XLP	-0.232	6	-2.3%	0.361	1.00	no	FLAT
PSCC	cluster_17	Mixed: VDC, FSTA, XLP	-0.232	6	-2.3%	0.085	1.00	no	LONG

Watchlist (imminent-buy clusters, p > 80%, not yet long)

ticker	cluster	theme	prob_buy	cluster_nss	basket_weight	basket_cor	sticky	etf_signal
ONEQ	cluster_3	Mixed: ONEQ, IGM, ILCG	1.00	1.000	0.527	0.99	no	LONG
VEA	cluster_1	Mixed: VEA, SPDW, SCHF	1.00	0.622	0.474	1.00	no	LONG
VTEB	cluster_16	Mixed: VTEB, MUB, CMF	1.00	0.244	0.458	0.99	no	FLAT
CXSE	cluster_13	Mixed: CXSE, GXC, MCHI	1.00	0.858	0.454	0.99	no	FLAT
IGOV	cluster_24	Mixed: IGOV, BWX, FXE	1.00	0.873	0.440	0.98	no	LONG
GNR	cluster_12	Mixed: GNR, MXI, GUNR	1.00	0.550	0.410	0.99	no	FLAT
EWJ	cluster_20	Mixed: EWJ, BBJP, HEWJ	1.00	0.947	0.410	1.00	no	LONG
DXJ	cluster_20	Mixed: EWJ, BBJP, HEWJ	1.00	0.947	0.405	1.00	no	LONG
FXL	cluster_3	Mixed: ONEQ, IGM, ILCG	1.00	1.000	0.376	0.99	no	FLAT
GSY	cluster_23	Mixed: GSY, FTSM, SHV	1.00	1.000	0.376	0.95	no	LONG
IYM	cluster_12	Mixed: GNR, MXI, GUNR	1.00	0.550	0.369	0.99	no	FLAT
MCHI	cluster_13	Mixed: CXSE, GXC, MCHI	1.00	0.858	0.320	0.99	no	FLAT
FXF	cluster_24	Mixed: IGOV, BWX, FXE	1.00	0.873	0.292	0.98	no	LONG
CMF	cluster_16	Mixed: VTEB, MUB, CMF	1.00	0.244	0.267	0.99	no	LONG
VGK	cluster_1	Mixed: VEA, SPDW, SCHF	1.00	0.622	0.242	1.00	no	FLAT
VYMI	cluster_1	Mixed: VEA, SPDW, SCHF	1.00	0.622	0.226	1.00	no	LONG
PICK	cluster_12	Mixed: GNR, MXI, GUNR	1.00	0.550	0.207	0.99	no	FLAT
CNYA	cluster_13	Mixed: CXSE, GXC, MCHI	1.00	0.858	0.207	0.99	no	LONG
FXB	cluster_24	Mixed: IGOV, BWX, FXE	1.00	0.873	0.203	0.98	no	LONG
HYD	cluster_16	Mixed: VTEB, MUB, CMF	1.00	0.244	0.172	0.99	no	LONG
VPL	cluster_20	Mixed: EWJ, BBJP, HEWJ	1.00	0.947	0.171	1.00	no	LONG
VNLA	cluster_23	Mixed: GSY, FTSM, SHV	1.00	1.000	0.139	0.95	no	LONG
NEAR	cluster_23	Mixed: GSY, FTSM, SHV	1.00	1.000	0.124	0.95	no	FLAT
ARKW	cluster_3	Mixed: ONEQ, IGM, ILCG	1.00	1.000	0.073	0.99	no	FLAT
ILF	cluster_18	Mixed: ILF, EWZ, ECH	0.98	0.072	0.400	0.96	no	FLAT
EPI	cluster_25	Mixed: INDA, EPI, PIN	0.98	0.084	0.256	1.00	no	FLAT
INDA	cluster_25	Mixed: INDA, EPI, PIN	0.98	0.084	0.250	1.00	no	FLAT
PIN	cluster_25	Mixed: INDA, EPI, PIN	0.98	0.084	0.249	1.00	no	FLAT
INDY	cluster_25	Mixed: INDA, EPI, PIN	0.98	0.084	0.246	1.00	no	FLAT
EIDO	cluster_18	Mixed: ILF, EWZ, ECH	0.98	0.072	0.200	0.96	no	FLAT

ECH	cluster_18	Mixed: ILF, EWZ, ECH	0.98	0.072	0.130	0.96	no	LONG
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Cluster Overview

change shows the signal-state shift vs the prior run. dNSS is today's NSS minus the prior run's.

	id	name	category	basket	n	signal	change	days	cum_return	nss	dNSS
3	cluster_3	Mixed: ONEQ, IGM, ILCG	Mixed	ONEQ+FXL+ARKW (cor=0.99)	63	FLAT	shifted			1.000	+0.585
8	cluster_8	Mixed: IEMG, EEM, SPEM	Mixed	IEMG+DEM+FEMS (cor=1.00)	25	LONG	shifted	21	9.9%	1.000	+0.595
22	cluster_22	Mixed: SRLN, BKLN, FTSL	Mixed	SRLN+LQDH+FL0T (cor=0.97)	9	LONG	shifted	24	1.2%	1.000	+0.186
23	cluster_23	Mixed: GSY, FTSM, SHV	Mixed	GSY+NEAR+VNLA (cor=0.95)	9	FLAT	shifted			1.000	+0.000
4	cluster_4	Mixed: IWB, ITOT, SCHB	Mixed	IWB+VT+IYC (cor=1.00)	61	LONG	shifted	20	5.9%	0.987	+1.183
20	cluster_20	Mixed: EWJ, BBJP, HEWJ	Mixed	EWJ+DXJ+VPL (cor=1.00)	9	FLAT	shifted			0.947	-0.014
6	cluster_6	Mixed: IWD, SCHV, VONV	Mixed	IWD+CFA+DGRO (cor=1.00)	54	LONG	shifted	20	2.6%	0.919	-0.070
21	cluster_21	Mixed: GSG, DBC, PDBC	Mixed	GSG+DBO+PDBC (cor=1.00)	9	LONG	shifted	84	47.8%	0.918	+0.102
24	cluster_24	Mixed: IGOV, BWX, FXE	Mixed	IGOV+FXF+FXB (cor=0.98)	6	FLAT	shifted			0.873	-0.127
13	cluster_13	Mixed: CXSE, GXC, MCHI	Mixed	CXSE+CNVA+MCHI (cor=0.99)	15	FLAT	shifted			0.858	+1.200
2	cluster_2	Mixed: VB, SCHM, IJH	Mixed	VB+PRN+RSPD (cor=1.00)	66	LONG	shifted	20	3.7%	0.849	+0.110
14	cluster_14	Mixed: SCHH, VNQ, FREL	Mixed	SCHH+RWR+INDS (cor=1.00)	14	LONG	shifted	21	4.2%	0.816	+0.852
10	cluster_10	Mixed: DVY, SPYD, SDOG	Mixed	DVY+AML+FDL (cor=0.99)	20	LONG	shifted	14	0.6%	0.787	+0.644
19	cluster_19	Mixed: TDTF, TIPX, TIP	Mixed	TIP+STPZ+SPIP (cor=1.00)	10	LONG	shifted	22	0.8%	0.787	+0.249
1	cluster_1	Mixed: VEA, SPDW, SCHF	Mixed	VEA+VGK+VYMI (cor=1.00)	82	FLAT	shifted			0.622	-0.378
12	cluster_12	Mixed: GNR, MXI, GUNR	Mixed	GNR+IYM+PICK (cor=0.99)	17	FLAT	shifted			0.550	+0.517
9	cluster_9	Mixed: JNK, HYG, HYL	Mixed	JNK+PFF+PCY (cor=0.99)	22	LONG	shifted	20	0.7%	0.489	+1.459
7	cluster_7	Mixed: VBR, MDYV, IVOV	Mixed	VBR+KBE+SPSM (cor=1.00)	48	LONG	shifted	21	2.9%	0.380	-0.619
16	cluster_16	Mixed: VTEB, MUB, CMF	Mixed	VTEB+HYD+CMF (cor=0.99)	13	FLAT	shifted			0.244	-0.051
25	cluster_25	Mixed: INDA, EPI, PIN	Mixed	INDA+INDY+EPI+PIN (cor=1.00)	4	FLAT	shifted			0.084	-0.916
18	cluster_18	Mixed: ILF, EWZ, ECH	Mixed	ILF+EIDO+ECH (cor=0.96)	11	FLAT	shifted			0.072	-0.138
11	cluster_11	Mixed: VHT, FHLC, FXH	Mixed	FXH+XBI+VHT (cor=0.99)	17	LONG	shifted	3	-2.0%	0.021	-0.188
5	cluster_5	Mixed: BND, AGG, SCHZ	Mixed	BND+BLV+BIV (cor=1.00)	58	LONG	shifted	0	0.0%	-0.177	-1.089
15	cluster_15	Mixed: DBP, GLTR, GDXJ	Mixed	GDXJ+GLTR+SIL (cor=1.00)	13	FLAT	shifted			-0.193	-0.964
17	cluster_17	Mixed: VDC, FSTA, XLP	Mixed	VDC+FUTY+PSCC (cor=1.00)	11	LONG	shifted	6	-2.3%	-0.232	-1.025
26	cluster_26	Mixed: UUP, SQQQ	Mixed	SQQQ+UUP (cor=1.00)	2	FLAT	shifted			-1.000	-1.791

Net Signal Strength (NSS)

NSS = P(buy crossover within 5 days) - P(sell crossover within 5 days). Range [-1, 1]; positive = bullish lean, negative = bearish lean.

ETF rows only (Cluster Overview already covers clusters), cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive 126d trailing return — i.e. would not be blocked by the strategy's entry filter.

Top 20 most bullish (NSS desc, abs-mom approved, cluster-deduped)

id	name	category	cluster	cluster_name	cluster_signal	prob_buy	prob_sell	nss
NXTG			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1	0.00	1.000
VLUE			cluster_6	Mixed: IWD, SCHV, VONV	LONG	1	0.00	1.000
IBDR			cluster_5	Mixed: BND, AGG, SCHZ	LONG	1	0.00	1.000
MINT			cluster_23	Mixed: GSY, FTSM, SHV	FLAT	1	0.00	1.000
FLTR			cluster_22	Mixed: SRLN, BKLN, FTSL	LONG	1	0.00	1.000
IHF			cluster_11	Mixed: VHT, FHLC, FXH	LONG	1	0.00	1.000
EWT			cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1	0.00	1.000
EWM			cluster_18	Mixed: ILF, EWZ, ECH	FLAT	1	0.00	1.000
FV			cluster_2	Mixed: VB, SCHM, IJH	LONG	1	0.00	1.000
EIS			cluster_4	Mixed: IWB, ITOT, SCHB	LONG	1	0.00	1.000
VPL			cluster_20	Mixed: EWJ, BBJP, HEWJ	FLAT	1	0.00	1.000
SLX			cluster_12	Mixed: GNR, MXI, GUNR	FLAT	1	0.00	1.000
FXV			cluster_24	Mixed: IGOV, BWX, FXE	FLAT	1	0.00	1.000
FTGC			cluster_21	Mixed: GSG, DBC, PDBC	LONG	1	0.00	1.000
XLP			cluster_17	Mixed: VDC, FSTA, XLP	LONG	1	0.00	0.999
NURE			cluster_14	Mixed: SCHH, VNQ, FREL	LONG	1	0.00	0.999
SCHD			cluster_10	Mixed: DVY, SPYD, SDOG	LONG	1	0.00	0.999
PFXF			cluster_9	Mixed: JNK, HYG, HYL	LONG	1	0.00	0.998
ENZL			cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	1	0.01	0.993
IWN			cluster_7	Mixed: VBR, MDYV, IVOV	LONG	1	0.01	0.987

Top 20 most bearish (NSS asc, cluster-deduped)

id	name	category	cluster	cluster_name	cluster_signal	prob_buy	prob_sell	nss
IHI			cluster_11	Mixed: VHT, FHLC, FXH	LONG	0.00	1.00	-1.000
SQQQ			cluster_26	Mixed: UUP, SQQQ	FLAT	0.00	1.00	-1.000
SPLV			cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.00	1.00	-1.000
EIDO			cluster_18	Mixed: ILF, EWZ, ECH	FLAT	0.00	1.00	-0.997
IDU			cluster_17	Mixed: VDC, FSTA, XLP	LONG	0.00	0.99	-0.992

LVHD	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.04	0.99	-0.952
PPA	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.00	0.93	-0.926
IEF	cluster_5	Mixed: BND, AGG, SCHZ	LONG	0.04	0.96	-0.926
MOO	cluster_12	Mixed: GNR, MXI, GUNR	FLAT	0.05	0.86	-0.816
GOAU	cluster_15	Mixed: DBP, GLTR, GDXJ	FLAT	0.10	0.88	-0.787
EFAV	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.22	0.94	-0.724
SHM	cluster_16	Mixed: VTEB, MUB, CMF	FLAT	0.26	0.96	-0.697
ESPO	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.40	0.97	-0.564
KIE	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.56	1.00	-0.436
CHIQ	cluster_13	Mixed: CXSE, GXC, MCHI	FLAT	0.57	0.90	-0.333
INDY	cluster_25	Mixed: INDA, EPI, PIN	FLAT	0.67	0.98	-0.306
IBUY	cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	0.87	0.98	-0.110
ANGL	cluster_9	Mixed: JNK, HYG, HYL	LONG	1.00	0.95	0.053
LTPZ	cluster_19	Mixed: TDTF, TIPX, TIP	LONG	0.87	0.58	0.292
FXE	cluster_24	Mixed: IGOV, BWX, FXE	FLAT	1.00	0.65	0.346

Cluster Details**Cluster 1 – Mixed: VEA, SPDW, SCHF****Members:** 82 **Var explained:** 0.839 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Unknown	82	1

Replication basket – VEA+VGK+VYMI (cor=1.00) basket lead at lag 1: -0.001

ticker	weight_ols	name	category	etf_signal
VEA	0.474			LONG
VGK	0.242			FLAT
VYMI	0.226			LONG

Members

	ticker	weight	name	category	etf_signal
2	EFA	0.0133			FLAT
3	VEA	0.0133			LONG
63	IEFA	0.0133			FLAT
64	SCHF	0.0133			LONG
65	SPDW	0.0133			LONG
21	DWM	0.0132			FLAT
34	TLTD	0.0132			FLAT
49	GSIE	0.0132			FLAT
52	ESGD	0.0132			FLAT
28	PXF	0.0131			LONG
38	ACWX	0.0131			LONG
39	CWI	0.0131			LONG
41	INTF	0.0131			FLAT
42	IQLT	0.0131			FLAT
44	QEFA	0.0131			FLAT
58	IEUR	0.0131			FLAT
59	IXUS	0.0131			LONG
60	VEU	0.0131			LONG

61	VGK	0.0131	FLAT
62	VXUS	0.0131	LONG
6	IEV	0.0130	FLAT
32	EFV	0.0130	FLAT
35	JPIN	0.0130	FLAT
46	RODM	0.0130	FLAT
47	VYMI	0.0130	LONG
48	FNDF	0.0130	LONG
56	SCHC	0.0130	LONG
75	FEU	0.0130	FLAT
24	IQDF	0.0129	LONG
31	SCZ	0.0129	LONG
33	FNDC	0.0129	LONG
57	VSS	0.0129	LONG
8	DLS	0.0128	LONG
19	EZU	0.0128	FLAT
43	IVLU	0.0128	FLAT
72	HFXI	0.0128	LONG
29	EFG	0.0127	FLAT
45	FEZ	0.0127	FLAT
54	VIGI	0.0127	LONG
4	FEP	0.0126	LONG
30	GWX	0.0126	LONG
40	IMTM	0.0126	LONG
1	EWG	0.0124	LONG
5	VIDI	0.0124	LONG
16	EWQ	0.0124	FLAT
20	IDV	0.0124	LONG
53	HDEF	0.0124	LONG
17	EWU	0.0123	LONG
50	IDLV	0.0123	LONG
79	IPKW	0.0123	LONG
82	EWI	0.0122	LONG
22	EPP	0.0121	FLAT
51	EFAV	0.0121	FLAT
23	EUFN	0.0120	FLAT
71	EWD	0.0120	LONG
77	EWN	0.0120	LONG
81	BBAX	0.0120	LONG
13	EWA	0.0119	FLAT
27	DWX	0.0119	LONG
12	HEZU	0.0118	FLAT
36	DBEF	0.0118	FLAT
37	HEFA	0.0118	FLAT
11	PID	0.0117	LONG

15	EWP	0.0116	FLAT
9	HEDJ	0.0114	FLAT
10	IHDG	0.0114	LONG
14	EWL	0.0114	LONG
18	EWL	0.0114	LONG
25	NFRA	0.0114	LONG
55	VNQI	0.0114	LONG
80	EWK	0.0113	FLAT
76	EWO	0.0112	LONG
7	RWX	0.0111	LONG
26	IGF	0.0108	LONG
70	EPOL	0.0104	FLAT
74	EZA	0.0099	LONG
69	FXA	0.0097	LONG
66	EBND	0.0096	LONG
67	EMLC	0.0096	LONG
73	ENZL	0.0091	LONG
68	EWV	0.0089	LONG
78	FXC	0.0084	LONG

Cluster 2 – Mixed: VB, SCHM, IJH

Members: 66 **Var explained:** 0.848 **Signal:** LONG since 2026-04-13 (20 days, 3.7%)

Category breakdown

category	n_members	abs_weight
Unknown	66	1

Replication basket – VB+PRN+RSPD (cor=1.00) basket lead at lag 1: -0.005

ticker	weight_ols	name	category	etf_signal
VB	0.687			LONG
PRN	0.158			LONG
RSPD	0.122			FLAT

Members

	ticker	weight	name	category	etf_signal
23	SCHM	0.0163			LONG
29	MDY	0.0163			LONG
37	IVOO	0.0163			LONG
42	IJH	0.0163			LONG
43	SPMD	0.0163			LONG
44	VB	0.0163			LONG
20	JHMM	0.0162			LONG
26	NUSC	0.0162			LONG
31	IJK	0.0162			LONG
33	IVOG	0.0162			LONG
34	MDYG	0.0162			LONG
45	SCHA	0.0162			LONG
27	SMLF	0.0161			LONG
41	VXF	0.0161			LONG
4	IWM	0.0160			LONG
30	IJT	0.0160			LONG
32	VI OG	0.0160			LONG
35	SLYG	0.0160			LONG
38	VTWO	0.0160			LONG
40	ISCG	0.0159			LONG
15	FXR	0.0158			LONG
28	IWO	0.0158			LONG
36	VTWG	0.0158			LONG
39	VBK	0.0158			LONG
54	PSC	0.0158			LONG
61	XMHQ	0.0158			LONG
14	IMCG	0.0157			LONG
21	VIS	0.0157			LONG
19	IYJ	0.0156			LONG
22	FIDU	0.0156			LONG
60	PAVE	0.0155			LONG
10	XMMO	0.0154			LONG
11	IWP	0.0153			FLAT
13	VOT	0.0153			FLAT
46	FXD	0.0153			LONG
62	VFMO	0.0153			LONG
66	XSMO	0.0153			LONG
1	XLI	0.0152			LONG
59	PSCI	0.0151			LONG
6	FTC	0.0150			LONG
24	DWAS	0.0150			LONG
25	IWC	0.0150			LONG
53	RSPD	0.0150			FLAT

7	PDP	0.0149	LONG
9	RPG	0.0148	LONG
18	FIW	0.0148	FLAT
48	FTXR	0.0148	LONG
5	FV	0.0147	LONG
63	PKB	0.0147	LONG
64	PRN	0.0146	LONG
16	PHO	0.0145	FLAT
8	FPX	0.0143	LONG
57	XTN	0.0143	FLAT
47	PEJ	0.0142	FLAT
65	PSCD	0.0142	FLAT
52	IYT	0.0141	LONG
2	XRT	0.0139	FLAT
58	QMOM	0.0138	LONG
12	MTUM	0.0137	LONG
56	XHB	0.0137	LONG
17	CGW	0.0135	FLAT
55	XAR	0.0135	FLAT
49	PPA	0.0133	FLAT
3	JETS	0.0127	LONG
50	ITA	0.0125	FLAT
51	ITB	0.0124	FLAT

Cluster 3 – Mixed: ONEQ, IGM, ILCG**Members:** 63 **Var explained:** 0.840 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Unknown	63	1

Replication basket – ONEQ+FXL+ARKW (cor=0.99) basket lead at lag 1: -0.007

ticker	weight_ols	name	category	etf_signal
ONEQ	0.527			LONG
FXL	0.376			FLAT

ARKW 0.073 FLAT

Members

	ticker	weight	name	category	etf_signal
14	QQQ	0.0170			LONG
35	IGM	0.0170			FLAT
40	ONEQ	0.0170			LONG
50	ILCG	0.0170			FLAT
29	XNTK	0.0169			FLAT
39	NULG	0.0169			FLAT
43	IWF	0.0169			LONG
45	VGT	0.0169			FLAT
47	FTEC	0.0169			FLAT
48	VONG	0.0169			LONG
52	SCHG	0.0169			FLAT
54	VUG	0.0169			FLAT
8	AIQ	0.0168			FLAT
18	QTEC	0.0168			FLAT
24	FXL	0.0168			FLAT
37	IYW	0.0168			FLAT
44	IVW	0.0168			LONG
51	IUSG	0.0168			LONG
17	QQEW	0.0167			FLAT
36	IXN	0.0167			LONG
42	IWY	0.0167			LONG
46	VOOG	0.0167			LONG
49	MGK	0.0167			FLAT
53	SPYG	0.0167			LONG
1	XLK	0.0166			FLAT
21	XT	0.0165			LONG
34	PWB	0.0165			LONG
55	SPHB	0.0162			LONG
20	TDIV	0.0160			LONG
28	FDN	0.0160			FLAT
60	FIVG	0.0160			FLAT
6	BOTZ	0.0159			LONG
7	ROBO	0.0159			LONG
63	QTUM	0.0159			LONG
27	PNQI	0.0158			FLAT
5	SKYY	0.0157			FLAT
2	SOXX	0.0156			LONG
3	SMH	0.0156			LONG

4	IGV	0.0155	FLAT
22	XSD	0.0155	LONG
38	CWB	0.0155	LONG
61	PBS	0.0155	FLAT
62	PTF	0.0155	LONG
13	ARKQ	0.0154	LONG
15	NXTG	0.0154	LONG
19	PSI	0.0153	LONG
57	PSCT	0.0153	LONG
26	IPO	0.0152	FLAT
31	FINX	0.0152	FLAT
33	QYLD	0.0152	LONG
59	FTXL	0.0152	LONG
12	ARKW	0.0151	FLAT
41	ICVT	0.0151	LONG
11	ARKK	0.0148	FLAT
25	CIBR	0.0148	FLAT
32	PTNQ	0.0148	LONG
23	IBUY	0.0147	FLAT
10	HACK	0.0146	FLAT
30	IPAY	0.0146	FLAT
16	PRNT	0.0144	FLAT
9	BLOK	0.0137	FLAT
56	PTLC	0.0128	LONG
58	BLCN	0.0123	FLAT

Cluster 4 – Mixed: IWB, ITOT, SCHB

Members: 61 **Var explained:** 0.892 **Signal:** LONG since 2026-04-13 (20 days, 5.9%)

Category breakdown

category	n_members	abs_weight
Unknown	61	1

Replication basket – IWB+VT+IYC (cor=1.00) basket lead at lag 1: -0.003

ticker	weight_ols	name	category	etf_signal
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IWB	0.636	LONG
VT	0.233	LONG
IYC	0.086	LONG

Members

	ticker	weight	name	category	etf_signal
37	IWB	0.0174			LONG
49	ITOT	0.0174			LONG
51	SCHB	0.0174			LONG
4	SPY	0.0173			LONG
12	URTH	0.0173			LONG
30	IWV	0.0173			LONG
31	IYY	0.0173			LONG
34	SPYX	0.0173			LONG
36	ESGU	0.0173			LONG
42	VTHR	0.0173			LONG
43	GSLC	0.0173			LONG
45	VONE	0.0173			LONG
47	VV	0.0173			LONG
48	ILCB	0.0173			LONG
50	IVV	0.0173			LONG
52	SCHX	0.0173			LONG
53	SPLG	0.0173			LONG
54	SPTM	0.0173			LONG
55	VOO	0.0173			LONG
56	VTI	0.0173			LONG
26	JHML	0.0172			LONG
29	SUSA	0.0172			LONG
38	IWL	0.0172			LONG
44	EPS	0.0172			LONG
46	MGC	0.0172			LONG
7	ACWI	0.0171			LONG
13	CRBN	0.0171			LONG
15	VT	0.0171			LONG
22	TILT	0.0171			LONG
28	DSI	0.0171			LONG
32	LRGF	0.0171			LONG
8	AOA	0.0170			LONG
14	SPGM	0.0170			LONG
33	OEF	0.0170			LONG
39	QUAL	0.0170			LONG
24	QDF	0.0168			LONG

40	QUS	0.0168	LONG
23	IOO	0.0167	LONG
25	CATH	0.0167	LONG
35	XLG	0.0167	FLAT
41	SPHQ	0.0166	LONG
11	AOR	0.0165	LONG
27	DGRW	0.0165	LONG
17	IYC	0.0162	LONG
60	OMFL	0.0161	LONG
21	FDIS	0.0159	LONG
18	VCR	0.0158	LONG
1	XLY	0.0156	LONG
10	AOM	0.0155	LONG
19	VOX	0.0153	LONG
20	FCOM	0.0153	LONG
2	XLC	0.0149	LONG
5	QAI	0.0149	LONG
59	RTH	0.0147	LONG
9	AOK	0.0142	LONG
61	IGN	0.0139	FLAT
57	IYZ	0.0137	FLAT
16	FPXI	0.0133	LONG
3	ESPO	0.0132	FLAT
58	EIS	0.0126	LONG
6	EWS	0.0121	LONG

Cluster 5 – Mixed: BND, AGG, SCHZ**Members:** 58 **Var explained:** 0.832 **Signal:** LONG since 2026-05-11 (0 days, 0.0%)**Category breakdown**

category	n_members	abs_weight
Unknown	58	1

Replication basket – BND+BLV+BIV (cor=1.00) basket lead at lag 1: 0.001

ticker	weight_ols	name	category	etf_signal
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BND	0.505	LONG
BLV	0.134	FLAT
BIV	0.219	FLAT

Members

	ticker	weight	name	category	etf_signal
21	BIV	0.0188			FLAT
26	AGG	0.0188			LONG
28	BND	0.0188			LONG
30	SCHZ	0.0188			LONG
20	IUSB	0.0187			LONG
31	SPAB	0.0187			LONG
2	IEF	0.0185			FLAT
17	IGIB	0.0184			LONG
24	VCIT	0.0184			LONG
7	FBND	0.0183			LONG
43	SPTI	0.0183			FLAT
47	VGIT	0.0183			FLAT
13	AGGY	0.0182			LONG
15	SPIB	0.0182			LONG
46	SCHR	0.0182			FLAT
6	GTO	0.0181			LONG
11	GVI	0.0181			LONG
40	IEI	0.0181			FLAT
12	QLTA	0.0180			LONG
27	USIG	0.0180			LONG
50	MBB	0.0180			LONG
51	VMBS	0.0180			LONG
3	LQD	0.0179			LONG
5	BOND	0.0179			LONG
52	SPMB	0.0179			LONG
19	ISTB	0.0178			FLAT
8	FLCO	0.0177			LONG
10	CORP	0.0177			LONG
22	BLV	0.0176			FLAT
41	TLH	0.0176			FLAT
45	GOVT	0.0176			FLAT
35	IGSB	0.0175			FLAT
37	VCSH	0.0175			FLAT
23	BSV	0.0174			LONG
49	GNMA	0.0174			LONG
18	IGLB	0.0172			LONG

44	SPTL	0.0172	FLAT
16	SPLB	0.0171	LONG
25	VCLT	0.0171	LONG
36	SLQD	0.0171	FLAT
48	VGLT	0.0171	FLAT
29	TOTL	0.0170	FLAT
1	TLT	0.0169	FLAT
39	AGZ	0.0166	LONG
9	BAB	0.0165	LONG
42	EDV	0.0163	FLAT
34	SPSB	0.0162	LONG
53	SHY	0.0162	FLAT
56	VGSH	0.0159	FLAT
57	BNDX	0.0158	LONG
33	BSCQ	0.0157	FLAT
55	SCHO	0.0156	FLAT
58	IAGG	0.0156	LONG
54	SPTS	0.0155	FLAT
14	IBDR	0.0152	FLAT
38	CMBS	0.0136	LONG
4	LMBS	0.0133	FLAT
32	LDUR	0.0120	LONG

Cluster 6 – Mixed: IWD, SCHV, VONV

Members: 54 **Var explained:** 0.922 **Signal:** LONG since 2026-04-13 (20 days, 2.6%)

Category breakdown

category	n_members	abs_weight
Unknown	54	1

Replication basket – IWD+CFA+DGRO (cor=1.00) basket lead at lag 1: 0.010

ticker	weight_ols	name	category	etf_signal
IWD	0.459			LONG
CFA	0.310			LONG
DGRO	0.222			LONG

Members

	ticker	weight	name	category	etf_signal
22	IWD	0.0192			LONG
2	RSP	0.0191			LONG
9	PRF	0.0191			LONG
29	VONV	0.0191			LONG
32	ILCV	0.0191			LONG
35	SCHV	0.0191			LONG
12	CFA	0.0190			LONG
18	FNDX	0.0190			LONG
24	IVE	0.0190			LONG
25	JPUS	0.0190			LONG
34	IUSV	0.0190			LONG
38	VTV	0.0190			LONG
45	DTD	0.0190			LONG
50	DGRO	0.0190			LONG
15	NULV	0.0189			LONG
20	IWX	0.0189			LONG
28	VOOV	0.0189			LONG
36	SPYV	0.0189			LONG
44	DLN	0.0189			LONG
5	FEX	0.0188			LONG
10	RWL	0.0188			LONG
30	MGV	0.0188			LONG
31	VYM	0.0188			LONG
39	IWS	0.0188			LONG
40	VOE	0.0188			LONG
42	PFM	0.0188			LONG
11	AIVL	0.0187			LONG
21	ONEV	0.0187			LONG
26	SIZE	0.0187			LONG
51	VIG	0.0187			LONG
53	JPME	0.0187			LONG
19	EQAL	0.0186			LONG
33	IMCB	0.0186			LONG
54	JVAL	0.0186			LONG
17	FDVV	0.0185			LONG
23	IWR	0.0185			LONG
37	VO	0.0185			LONG
47	DIA	0.0184			LONG
4	PKW	0.0183			FLAT
14	NOBL	0.0182			FLAT
6	PWV	0.0181			LONG

7	RDVY	0.0181	LONG
3	FVD	0.0180	FLAT
16	SDY	0.0180	LONG
27	VLUE	0.0180	LONG
43	OUSA	0.0180	LONG
41	FTCS	0.0179	FLAT
49	LGLV	0.0179	FLAT
13	CFO	0.0178	LONG
8	MOAT	0.0177	LONG
48	USMV	0.0176	LONG
52	ETHO	0.0176	LONG
1	ACWV	0.0168	LONG
46	SPLV	0.0159	FLAT

Cluster 7 – Mixed: VBR, MDYV, IVOV**Members:** 48 **Var explained:** 0.880 **Signal:** LONG since 2026-04-10 (21 days, 2.9%)**Category breakdown**

category	n_members	abs_weight
Unknown	48	1

Replication basket – VBR+KBE+SPSM (cor=1.00) basket lead at lag 1: -0.000

ticker	weight_ols	name	category	etf_signal
VBR	0.557			LONG
KBE	0.164			LONG
SPSM	0.222			LONG

Members

	ticker	weight	name	category	etf_signal
37	VBR	0.0220			LONG
28	IJJ	0.0219			LONG
30	IVOV	0.0219			LONG

31	MDYV	0.0219	LONG
21	EZM	0.0218	LONG
23	FNDA	0.0218	LONG
8	FYX	0.0217	LONG
10	VIOO	0.0217	LONG
11	IJR	0.0217	LONG
12	SPSM	0.0217	LONG
19	DON	0.0217	LONG
26	IWN	0.0217	LONG
29	IJS	0.0217	LONG
32	SLYV	0.0217	LONG
34	VTWV	0.0217	LONG
47	SDVY	0.0217	LONG
7	FNX	0.0216	LONG
18	DES	0.0216	LONG
20	EES	0.0216	LONG
33	VIOV	0.0216	LONG
9	PRFZ	0.0215	LONG
46	FYT	0.0215	LONG
41	RWK	0.0214	LONG
38	RWJ	0.0213	LONG
48	ISMD	0.0213	LONG
39	RFV	0.0212	LONG
42	OMFS	0.0210	LONG
13	FXO	0.0209	FLAT
43	RZV	0.0209	LONG
6	REGL	0.0207	LONG
17	SMDV	0.0206	LONG
25	XSLV	0.0204	LONG
35	VFH	0.0204	FLAT
36	FNCL	0.0204	FLAT
27	SMMV	0.0203	LONG
3	KBE	0.0202	LONG
15	IYF	0.0202	FLAT
45	KCE	0.0202	FLAT
16	IYG	0.0201	FLAT
24	XMLV	0.0198	FLAT
1	XLF	0.0197	FLAT
22	KBWB	0.0197	FLAT
2	KRE	0.0196	LONG
14	IAT	0.0194	LONG
44	IAI	0.0189	FLAT
40	PSP	0.0181	FLAT
4	KIE	0.0177	FLAT
5	REM	0.0173	LONG

Cluster 8 – Mixed: IEMG, EEM, SPEM**Members:** 25 **Var explained:** 0.875 **Signal:** LONG since 2026-04-10 (21 days, 9.9%)**Category breakdown**

category	n_members	abs_weight
Unknown	25	1

Replication basket – IEMG+DEM+FEMS (cor=1.00) basket lead at lag 1: -0.023

ticker	weight_ols	name	category	etf_signal
IEMG	0.754			LONG
DEM	0.168			LONG
FEMS	0.067			FLAT

Members

	ticker	weight	name	category	etf_signal
21	IEMG	0.0425			LONG
2	EEM	0.0424			LONG
16	GEM	0.0423			LONG
23	SPEM	0.0423			LONG
3	VWO	0.0422			LONG
20	ESGE	0.0422			LONG
22	SCHE	0.0421			LONG
15	EMGF	0.0419			LONG
18	XSOE	0.0419			LONG
5	AAXJ	0.0418			LONG
11	EEMA	0.0413			LONG
14	GMF	0.0411			LONG
12	PXH	0.0410			LONG
17	FNDE	0.0410			LONG
19	EEMV	0.0404			LONG
8	DGS	0.0402			LONG
10	AIA	0.0401			LONG
7	DEM	0.0396			LONG
6	EWX	0.0390			LONG
4	FEM	0.0385			LONG
13	DVYE	0.0368			FLAT

9	EWT	0.0357	LONG
24	SDIV	0.0348	LONG
25	FEMS	0.0347	FLAT
1	EWY	0.0341	LONG

Cluster 9 – Mixed: JNK, HYG, HYL

Members: 22 **Var explained:** 0.772 **Signal:** LONG since 2026-04-13 (20 days, 0.7%)

Category breakdown

category	n_members	abs_weight
Unknown	22	1

Replication basket – JNK+PFF+PCY (cor=0.99) basket lead at lag 1: 0.033

ticker	weight_ols	name	category	etf_signal
JNK	0.487			LONG
PFF	0.257			LONG
PCY	0.115			FLAT

Members

	ticker	weight	name	category	etf_signal
14	JNK	0.0502			LONG
1	HYG	0.0501			LONG
20	HYLB	0.0500			LONG
19	BBHY	0.0499			LONG
17	SHYG	0.0497			LONG
15	SJNK	0.0496			LONG
8	PHB	0.0493			LONG
16	ANGL	0.0492			LONG
18	FALN	0.0492			LONG
6	HYS	0.0488			LONG
10	PFF	0.0453			LONG
3	HYLS	0.0450			LONG

2	EMB	0.0444	FLAT
22	VWOB	0.0443	FLAT
12	PFXF	0.0438	LONG
21	PCY	0.0435	FLAT
7	PGX	0.0421	LONG
11	PSK	0.0416	FLAT
4	FPE	0.0410	LONG
5	PGF	0.0407	FLAT
13	HYEM	0.0373	LONG
9	VRP	0.0348	LONG

Cluster 10 – Mixed: DVY, SPYD, SDOG

Members: 20 **Var explained:** 0.832 **Signal:** LONG since 2026-04-21 (14 days, 0.6%)

Category breakdown

category	n_members	abs_weight
Unknown	20	1

Replication basket – DVY+AML+FDL (cor=0.99) basket lead at lag 1: 0.029

ticker	weight_ols	name	category	etf_signal
DVY	0.571			LONG
AML	0.129			LONG
FDL	0.263			FLAT

Members

	ticker	weight	name	category	etf_signal
15	DVY	0.0538			LONG
20	SPYD	0.0535			LONG
14	SDOG	0.0534			LONG
6	DHS	0.0533			LONG
19	ONEY	0.0528			LONG
11	FTA	0.0526			FLAT

5	FDL	0.0524	FLAT
18	SPHD	0.0524	LONG
10	SCHD	0.0521	LONG
13	DIV	0.0517	LONG
16	RDIV	0.0516	LONG
7	CDC	0.0515	LONG
17	RPV	0.0511	LONG
12	PEY	0.0510	LONG
9	HDV	0.0508	LONG
8	LVHD	0.0495	FLAT
1	EMLP	0.0472	LONG
4	MLPX	0.0419	LONG
2	AMLPA	0.0393	LONG
3	MLPA	0.0383	LONG

Cluster 11 – Mixed: VHT, FHLC, FXH

Members: 17 **Var explained:** 0.764 **Signal:** LONG since 2026-05-06 (3 days, -2.0%)

Category breakdown

category	n_members	abs_weight
Unknown	17	1

Replication basket – FXH+XBI+VHT (cor=0.99) basket lead at lag 1: -0.011

ticker	weight_ols	name	category	etf_signal
FXH	0.363			FLAT
XBI	0.185			FLAT
VHT	0.472			FLAT

Members

	ticker	weight	name	category	etf_signal
15	VHT	0.0653			FLAT
16	FHLC	0.0652			FLAT

9	FXH	0.0645	FLAT
14	RSPH	0.0639	FLAT
13	IYH	0.0637	FLAT
10	IXJ	0.0622	FLAT
1	XLV	0.0620	FLAT
3	IBB	0.0605	LONG
6	FBT	0.0595	FLAT
8	PSCH	0.0581	LONG
7	XHE	0.0574	FLAT
12	IHI	0.0573	FLAT
5	PTH	0.0545	FLAT
17	PPH	0.0542	FLAT
4	XBI	0.0535	FLAT
2	ARKG	0.0511	FLAT
11	IHF	0.0470	FLAT

Cluster 12 – Mixed: GNR, MXI, GUNR**Members:** 17 **Var explained:** 0.756 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Unknown	17	1

Replication basket – GNR+IYM+PICK (cor=0.99) basket lead at lag 1: 0.041

ticker	weight_ols	name	category	etf_signal
GNR	0.410			FLAT
IYM	0.369			FLAT
PICK	0.207			FLAT

Members

	ticker	weight	name	category	etf_signal
11	GNR	0.0653			FLAT

6	MXI	0.0648	FLAT
10	GUNR	0.0643	FLAT
7	IYM	0.0641	FLAT
12	PICK	0.0627	FLAT
9	VAW	0.0622	FLAT
17	NANR	0.0614	LONG
1	XLB	0.0612	FLAT
8	RSPM	0.0611	FLAT
13	SLX	0.0606	LONG
4	XME	0.0605	LONG
5	MOO	0.0589	LONG
3	COPX	0.0587	LONG
16	IGE	0.0571	LONG
14	NORW	0.0519	LONG
2	URA	0.0438	LONG
15	DBB	0.0414	LONG

Cluster 13 – Mixed: CXSE, GXC, MCHI**Members:** 15 **Var explained:** 0.804 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Unknown	15	1

Replication basket – CXSE+CNYY+MCHI (cor=0.99) basket lead at lag 1: -0.040

ticker	weight_ols	name	category	etf_signal
CXSE	0.454			FLAT
CNYY	0.207			LONG
MCHI	0.320			FLAT

Members

ticker	weight	name	category	etf_signal
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11	CXSE	0.0736	FLAT
9	GXC	0.0735	FLAT
1	MCHI	0.0729	FLAT
5	CQQQ	0.0712	FLAT
2	FXI	0.0711	FLAT
7	CHIQ	0.0706	FLAT
4	KWEB	0.0679	FLAT
6	ASHR	0.0669	LONG
8	CNYA	0.0661	LONG
13	KGRN	0.0651	FLAT
3	EMQQ	0.0642	FLAT
14	CNXT	0.0595	LONG
10	KBA	0.0594	LONG
12	EWI	0.0591	LONG
15	KURE	0.0587	FLAT

Cluster 14 – Mixed: SCHH, VNQ, FREL**Members:** 14 **Var explained:** 0.962 **Signal:** LONG since 2026-04-10 (21 days, 4.2%)**Category breakdown**

category	n_members	abs_weight
Unknown	14	1

Replication basket – SCHH+RWR+INDS (cor=1.00) basket lead at lag 1: -0.001

ticker	weight_ols	name	category	etf_signal
SCHH	0.541			LONG
RWR	0.336			LONG
INDS	0.100			LONG

Members

	ticker	weight	name	category	etf_signal
8	VNQ	0.0725			LONG

9	FREL	0.0725	LONG
11	SCHH	0.0725	LONG
2	IYR	0.0724	LONG
6	RWR	0.0723	LONG
10	USRT	0.0723	LONG
13	PSR	0.0721	LONG
1	XLRE	0.0719	LONG
5	ICF	0.0719	LONG
7	REET	0.0715	LONG
3	RWO	0.0711	LONG
4	REZ	0.0698	LONG
14	NURE	0.0689	LONG
12	INDS	0.0683	LONG

Cluster 15 – Mixed: DBP, GLTR, GDXJ**Members:** 13 **Var explained:** 0.831 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Unknown	13	1

Replication basket – GDXJ+GLTR+SIL (cor=1.00) basket lead at lag 1: 0.002

ticker	weight_ols	name	category	etf_signal
GDXJ	0.218			FLAT
GLTR	0.601			FLAT
SIL	0.168			LONG

Members

	ticker	weight	name	category	etf_signal
12	DBP	0.0814			FLAT
7	GLTR	0.0810			FLAT
2	GDXJ	0.0786			FLAT

1	GDX	0.0785	FLAT
3	GLD	0.0781	FLAT
10	IAU	0.0781	FLAT
11	SGOL	0.0779	FLAT
13	GOAU	0.0777	FLAT
6	SIL	0.0776	LONG
5	SILJ	0.0765	LONG
4	SLV	0.0763	LONG
9	SIVR	0.0761	LONG
8	PPLT	0.0622	LONG

Cluster 16 – Mixed: VTEB, MUB, CMF**Members:** 13 **Var explained:** 0.819 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Unknown	13	1

Replication basket – VTEB+HYD+CMF (cor=0.99) basket lead at lag 1: 0.188

ticker	weight_ols	name	category	etf_signal
VTEB	0.458			FLAT
HYD	0.172			LONG
CMF	0.267			LONG

Members

	ticker	weight	name	category	etf_signal
13	VTEB	0.0817			FLAT
11	MUB	0.0813			LONG
6	CMF	0.0801			LONG
9	TFI	0.0801			LONG
8	ITM	0.0800			LONG
1	FMB	0.0796			LONG

7	NYF	0.0793	LONG
5	PZA	0.0780	LONG
4	MUNI	0.0769	FLAT
2	HYD	0.0748	LONG
3	HYMB	0.0720	LONG
12	SUB	0.0694	LONG
10	SHM	0.0668	FLAT

Cluster 17 – Mixed: VDC, FSTA, XLP

Members: 11 **Var explained:** 0.748 **Signal:** LONG since 2026-05-01 (6 days, -2.3%)

Category breakdown

category	n_members	abs_weight
Unknown	11	1

Replication basket – VDC+FUTY+PSCC (cor=1.00) basket lead at lag 1: -0.008

ticker	weight_ols	name	category	etf_signal
VDC	0.527			LONG
FUTY	0.361			FLAT
PSCC	0.085			LONG

Members

	ticker	weight	name	category	etf_signal
5	VDC	0.0983			LONG
6	FSTA	0.0983			LONG
1	XLP	0.0973			LONG
4	IYK	0.0947			LONG
3	KXI	0.0944			LONG
11	PBJ	0.0897			FLAT
7	IDU	0.0892			FLAT
8	VPU	0.0880			FLAT
9	FUTY	0.0880			FLAT

2	XLU	0.0875	FLAT
10	PSCC	0.0746	LONG

Cluster 18 – Mixed: ILF, EWZ, ECH**Members:** 11 **Var explained:** 0.325 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Unknown	11	1

Replication basket – ILF+EIDO+ECH (cor=0.96) basket lead at lag 1: 0.018

ticker	weight_ols	name	category	etf_signal
ILF	0.40			FLAT
EIDO	0.20			FLAT
ECH	0.13			LONG

Members

	ticker	weight	name	category	etf_signal
4	ILF	0.1565			FLAT
1	EWZ	0.1432			FLAT
3	ECH	0.1273			LONG
8	EWM	0.1242			LONG
6	EIDO	0.1137			FLAT
7	EPHE	0.1051			FLAT
11	VNM	0.0792			FLAT
2	DBA	0.0598			FLAT
10	TUR	0.0575			LONG
9	RSX	0.0189			LONG
5	UNG	0.0147			FLAT

Cluster 19 – Mixed: TDTF, TIPX, TIP**Members:** 10 **Var explained:** 0.908 **Signal:** LONG since 2026-04-09 (22 days, 0.8%)**Category breakdown**

category	n_members	abs_weight
Unknown	10	1

Replication basket – TIP+STPZ+SPIP (cor=1.00) basket lead at lag 1: -0.019

ticker	weight_ols	name	category	etf_signal
TIP	0.627			LONG
STPZ	0.194			LONG
SPIP	0.136			LONG

Members

	ticker	weight	name	category	etf_signal
4	TDTF	0.1033			LONG
6	TIPX	0.1032			LONG
3	TIP	0.1028			LONG
8	SCHP	0.1023			LONG
5	TDTT	0.1007			LONG
9	STIP	0.0999			FLAT
10	VTIP	0.0998			FLAT
7	SPIP	0.0997			LONG
2	STPZ	0.0995			LONG
1	LTPZ	0.0886			FLAT

Cluster 20 – Mixed: EWJ, BBJP, HEWJ**Members:** 9 **Var explained:** 0.888 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Unknown	9	1

Replication basket – EWJ+DXJ+VPL (cor=1.00) basket lead at lag 1: -0.032

ticker	weight_ols	name	category	etf_signal
EWJ	0.410			LONG
DXJ	0.405			LONG
VPL	0.171			LONG

Members

	ticker	weight	name	category	etf_signal
1	EWJ	0.1141			LONG
8	BBJP	0.1141			LONG
9	HEWJ	0.1115			LONG
6	DBJP	0.1114			LONG
3	IPAC	0.1113			FLAT
5	FLJH	0.1106			FLAT
2	DXJ	0.1097			LONG
4	VPL	0.1091			LONG
7	EWJV	0.1082			LONG

Cluster 21 – Mixed: GSG, DBC, PDBC

Members: 9 **Var explained:** 0.913 **Signal:** LONG since 2026-01-08 (84 days, 47.8%)

Category breakdown

category	n_members	abs_weight
Unknown	9	1

Replication basket – GSG+DBO+PDBC (cor=1.00) basket lead at lag 1: 0.057

ticker	weight_ols	name	category	etf_signal
GSG	0.478			FLAT
DBO	0.179			FLAT
PDBC	0.353			LONG

Members

	ticker	weight	name	category	etf_signal
5	GSG	0.1151			FLAT
2	DBC	0.1149			LONG
7	PDBC	0.1149			LONG
8	COMT	0.1148			FLAT
9	DBE	0.1119			FLAT
1	USO	0.1098			FLAT
4	DBO	0.1085			FLAT
6	DJP	0.1051			FLAT
3	FTGC	0.1050			FLAT

Cluster 22 – Mixed: SRLN, BKLN, FTSL

Members: 9 **Var explained:** 0.560 **Signal:** LONG since 2026-04-07 (24 days, 1.2%)

Category breakdown

category	n_members	abs_weight
Unknown	9	1

Replication basket – SRLN+LQDH+FLOT (cor=0.97) basket lead at lag 1: -0.033

ticker	weight_ols	name	category	etf_signal
SRLN	0.346			LONG
LQDH	0.319			LONG
FLOT	0.322			FLAT

Members

	ticker	weight	name	category	etf_signal
2	SRLN	0.1249			LONG
3	BKLN	0.1213			LONG
1	FTSL	0.1170			LONG
5	LQDH	0.1145			LONG
7	FLOT	0.1081			FLAT
6	IGBH	0.1057			LONG
4	IGHG	0.1043			LONG
8	FLRN	0.1035			FLAT
9	FLTR	0.1008			FLAT

Cluster 23 – Mixed: GSY, FTSM, SHV**Members:** 9 **Var explained:** 0.457 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Unknown	9	1

Replication basket – GSY+NEAR+VNLA (cor=0.95) basket lead at lag 1: 0.026

ticker	weight_ols	name	category	etf_signal
GSY	0.376			LONG
NEAR	0.124			FLAT
VNLA	0.139			LONG

Members

	ticker	weight	name	category	etf_signal
5	GSY	0.1359			LONG
3	FTSM	0.1286			FLAT
7	SHV	0.1261			FLAT
6	ICSH	0.1244			FLAT
4	NEAR	0.1090			FLAT

2	MINT	0.1062	FLAT
1	VNLA	0.1032	LONG
8	BIL	0.0879	FLAT
9	GBIL	0.0787	FLAT

Cluster 24 – Mixed: IGOV, BWX, FXE**Members:** 6 **Var explained:** 0.742 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Unknown	6	1

Replication basket – IGOV+FXF+FXB (cor=0.98) basket lead at lag 1: 0.036

ticker	weight_ols	name	category	etf_signal
IGOV	0.440			LONG
FXF	0.292			LONG
FXB	0.203			LONG

Members

	ticker	weight	name	category	etf_signal
2	IGOV	0.1801			LONG
1	BWX	0.1789			LONG
4	FXE	0.1701			FLAT
5	FXF	0.1647			LONG
3	FXB	0.1609			LONG
6	FXY	0.1453			FLAT

Cluster 25 – Mixed: INDA, EPI, PIN**Members:** 4 **Var explained:** 0.964 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Unknown	4	1

Replication basket – INDA+INDY+EPI+PIN (cor=1.00) basket lead at lag 1: -0.071

ticker	weight_ols	name	category	etf_signal
INDA	0.250			FLAT
INDY	0.246			FLAT
EPI	0.256			FLAT
PIN	0.249			FLAT

Members

	ticker	weight	name	category	etf_signal
1	INDA	0.2527			FLAT
3	EPI	0.2493			FLAT
4	PIN	0.2492			FLAT
2	INDY	0.2489			FLAT

Cluster 26 – Mixed: UUP, SQQQ**Members:** 2 **Var explained:** 0.624 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Unknown	2	1

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.017

ticker	weight_ols	name	category	etf_signal
SQQQ	0.507			FLAT
UUP	0.496			LONG

Members

ticker	weight	name	category	etf_signal
SQQQ	0.5			FLAT
UUP	0.5			LONG

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	name	category	cluster	cluster_name	cluster_signal	cluster_nss	entry_date	days	cum_return	prob_sell	nss
GSY			cluster_23	Mixed: GSY, FTSM, SHV	FLAT	1.000	2022-11-11	874	20.4%	0.00	1.000
VNLA			cluster_23	Mixed: GSY, FTSM, SHV	FLAT	1.000	2026-04-02	26	0.5%	0.03	0.975
SRLN			cluster_22	Mixed: SRLN, BKLN, FT...	LONG	1.000	2026-04-07	24	1.9%	0.00	1.000
FTSL			cluster_22	Mixed: SRLN, BKLN, FT...	LONG	1.000	2026-04-06	25	1.4%	0.00	0.999
BKLN			cluster_22	Mixed: SRLN, BKLN, FT...	LONG	1.000	2026-04-06	25	1.3%	0.00	0.999
LQDH			cluster_22	Mixed: SRLN, BKLN, FT...	LONG	1.000	2026-04-06	25	1.3%	0.01	0.990
IGBH			cluster_22	Mixed: SRLN, BKLN, FT...	LONG	1.000	2026-04-08	23	1.7%	0.01	0.990
IGHG			cluster_22	Mixed: SRLN, BKLN, FT...	LONG	1.000	2026-04-07	24	1.1%	0.11	0.889
XSD			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-09	22	58.0%	0.00	1.000
NXTG			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-08	23	21.7%	0.00	1.000
FTXL			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-07	24	59.1%	0.00	1.000
PTNQ			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-17	16	9.9%	0.00	1.000
PSI			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-07	24	56.9%	0.00	1.000
SOXX			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-08	23	43.8%	0.00	1.000
SMH			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-08	23	36.3%	0.00	1.000
IXN			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-10	21	24.5%	0.00	1.000
PSCT			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-07	24	30.0%	0.00	1.000
QQQ			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-10	21	16.7%	0.00	1.000
QTUM			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-10	21	26.7%	0.00	1.000
PTLC			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-20	15	4.3%	0.00	1.000
QYLD			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-08	23	4.3%	0.00	1.000
PTF			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-07	24	35.1%	0.00	1.000
BOTZ			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-16	17	12.5%	0.00	1.000
XT			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-13	20	10.8%	0.00	1.000
TDIV			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-13	20	14.3%	0.00	1.000
ROBO			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-14	19	14.2%	0.00	1.000
ONEQ			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-13	20	13.4%	0.00	1.000
ICVT			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-08	23	10.4%	0.00	1.000
PWB			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-10	21	13.7%	0.00	1.000
SPHB			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-10	21	16.4%	0.00	1.000
CWB			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-08	23	10.1%	0.00	1.000
IWV			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-13	20	11.5%	0.00	0.999
SPYG			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-13	20	11.5%	0.00	0.999
VOOG			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-13	20	11.5%	0.00	0.999
IUSG			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-13	20	11.2%	0.00	0.999
IWY			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-14	19	7.7%	0.01	0.988
VONG			cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-14	19	7.3%	0.01	0.985

IWF	cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-14	19	7.3%	0.02	0.985
ARKQ	cluster_3	Mixed: ONEQ, IGM, ILCG	FLAT	1.000	2026-04-14	19	11.1%	0.06	0.941
EWT	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-08	23	27.2%	0.00	1.000
EWY	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-09	22	38.5%	0.00	1.000
AIA	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-10	21	21.0%	0.00	1.000
AAXJ	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-13	20	13.7%	0.00	1.000
EEMA	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-13	20	13.4%	0.00	1.000
EMGF	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-10	21	13.3%	0.00	1.000
XSOE	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-10	21	12.5%	0.00	1.000
EEM	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-10	21	12.1%	0.00	1.000
IEMG	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-10	21	11.7%	0.00	1.000
ESGE	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-13	20	11.2%	0.00	1.000
GEM	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-10	21	11.3%	0.00	1.000
GMF	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-14	19	6.1%	0.00	1.000
DEM	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-09	22	7.0%	0.00	1.000
EEMV	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-10	21	8.5%	0.00	1.000
PXH	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-10	21	5.7%	0.01	0.987
SPEM	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-13	20	5.7%	0.02	0.981
VVO	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-13	20	5.6%	0.02	0.975
SCHE	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-13	20	5.2%	0.03	0.974
EWX	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-10	21	8.5%	0.03	0.969
FNDE	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-10	21	4.9%	0.03	0.966
FEM	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-07	24	11.5%	0.09	0.914
DGS	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-10	21	4.0%	0.15	0.853
SDIV	cluster_8	Mixed: IEMG, EEM, SPEM	LONG	1.000	2026-04-09	22	1.2%	0.46	0.537
EIS	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-10	21	8.4%	0.00	1.000
DSI	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	8.6%	0.00	1.000
IOO	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	9.1%	0.00	1.000
QAI	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-08	23	4.0%	0.00	1.000
OEF	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	9.5%	0.00	1.000
SUSA	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	8.0%	0.00	1.000
OMFL	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-10	21	7.2%	0.00	1.000
ESGU	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	8.1%	0.00	1.000
MGC	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	8.7%	0.00	1.000
IVV	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.7%	0.00	0.999
SPY	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.8%	0.00	0.999
VOO	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.7%	0.00	0.999
VV	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.7%	0.00	0.999
IWL	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	8.2%	0.00	0.999
ILCB	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.7%	0.00	0.999
SPTM	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.4%	0.00	0.999
IYY	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.5%	0.00	0.999
EPS	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-10	21	8.1%	0.00	0.999
SCHB	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.4%	0.00	0.999
ITOT	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.4%	0.00	0.999

SCHX	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.5%	0.00	0.999
VTI	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.4%	0.00	0.999
SPYX	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.8%	0.00	0.999
IWB	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.2%	0.00	0.998
SPHQ	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-10	21	5.4%	0.00	0.998
IWV	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.2%	0.00	0.998
VTHR	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.3%	0.00	0.998
VONE	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.2%	0.00	0.998
CATH	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	7.3%	0.00	0.997
TILT	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	6.0%	0.00	0.996
ACWI	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	6.2%	0.01	0.995
JHML	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	5.6%	0.01	0.994
SPGM	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-10	21	7.0%	0.01	0.994
VT	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	6.1%	0.01	0.993
LRGF	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	6.4%	0.01	0.993
QDF	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	5.3%	0.01	0.992
CRBN	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	6.2%	0.01	0.989
AOA	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	4.8%	0.01	0.989
GSLC	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	6.2%	0.01	0.988
URTH	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	5.7%	0.01	0.986
AOR	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	3.6%	0.02	0.980
AOM	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	2.4%	0.05	0.952
QUS	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	2.5%	0.06	0.941
FPXI	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-10	21	7.3%	0.07	0.934
AOK	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	1.8%	0.07	0.928
QUAL	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	3.6%	0.08	0.924
DGRW	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	3.2%	0.11	0.890
XLY	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-15	18	1.0%	0.32	0.679
FDIS	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	2.4%	0.34	0.657
VCR	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	2.4%	0.35	0.647
VOX	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	2.3%	0.60	0.400
FCOM	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20	2.1%	0.62	0.377
XLC	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	-0.8%	0.74	0.235
IYC	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-14	19	-1.0%	0.82	0.180
EWS	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-07	24	1.9%	0.83	0.160
RTH	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-10	21	-0.0%	0.93	0.067
SPLG	cluster_4	Mixed: IWB, ITOT, SCHB	LONG	0.987	2026-04-13	20			
VPL	cluster_20	Mixed: EWJ, BBJP, HEWJ	FLAT	0.947	2026-04-10	21	10.1%	0.00	1.000
EWJ	cluster_20	Mixed: EWJ, BBJP, HEWJ	FLAT	0.947	2026-04-13	20	4.4%	0.03	0.974
BBJP	cluster_20	Mixed: EWJ, BBJP, HEWJ	FLAT	0.947	2026-04-30	7	3.4%	0.03	0.971
HEWJ	cluster_20	Mixed: EWJ, BBJP, HEWJ	FLAT	0.947	2026-04-08	23	2.7%	0.09	0.914
DBJP	cluster_20	Mixed: EWJ, BBJP, HEWJ	FLAT	0.947	2026-04-08	23	2.5%	0.13	0.870
EWJV	cluster_20	Mixed: EWJ, BBJP, HEWJ	FLAT	0.947	2026-05-06	3	-0.1%	0.17	0.825
DXJ	cluster_20	Mixed: EWJ, BBJP, HEWJ	FLAT	0.947	2026-05-01	6	4.2%	0.23	0.773
VLUE	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-09	22	21.2%	0.00	1.000

FNDX	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-09	22	5.6%	0.00	1.000
PRF	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-09	22	6.4%	0.00	1.000
IWX	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	6.3%	0.00	1.000
NULV	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	6.7%	0.00	1.000
JVAL	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	9.1%	0.00	1.000
IWD	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	5.8%	0.00	1.000
VONV	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	5.8%	0.00	1.000
MGV	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	3.5%	0.00	1.000
ILCV	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	4.0%	0.00	0.999
VOOV	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	3.3%	0.00	0.999
SPYV	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	3.2%	0.00	0.998
IVE	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	3.2%	0.00	0.998
SCHV	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	4.4%	0.00	0.998
IUSV	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	3.1%	0.00	0.997
RWL	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	4.8%	0.01	0.995
VTV	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	2.8%	0.01	0.993
IMCB	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	5.5%	0.04	0.959
FEX	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-09	22	4.2%	0.06	0.940
IWS	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-09	22	4.2%	0.06	0.937
IWR	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	4.9%	0.08	0.924
ETHO	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-09	22	5.6%	0.11	0.889
VO	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	3.3%	0.11	0.886
SIZE	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	3.4%	0.12	0.885
DTD	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	2.5%	0.12	0.881
AIVL	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	3.2%	0.12	0.881
DLN	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	2.6%	0.12	0.879
RSP	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	2.2%	0.20	0.805
VIG	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	2.0%	0.25	0.751
PFM	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	1.9%	0.26	0.739
FDVV	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	2.4%	0.27	0.726
DIA	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	2.4%	0.29	0.709
JPME	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-07	24	3.9%	0.34	0.659
VYM	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	2.5%	0.38	0.618
OUSA	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-20	15	-0.9%	0.47	0.528
DGRO	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	1.2%	0.54	0.463
MOAT	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-05-06	3	-0.9%	0.61	0.384
PWV	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-02	26	3.6%	0.62	0.379
RDVY	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-10	21	3.0%	0.70	0.302
CFO	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	0.7%	0.72	0.278
SDY	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-05-01	6	-0.9%	0.71	0.277
CFA	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-13	20	0.7%	0.73	0.272
USMV	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-05-05	4	-0.4%	0.64	0.245
EQAL	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-08	23	1.5%	0.76	0.238
ACWV	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-05-06	3	-0.4%	0.66	0.212
ONEV	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-14	19	0.3%	0.85	0.152

JPUS	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-08	23	0.9%	0.85	0.150
VOE	cluster_6	Mixed: IWD, SCHV, VONV	LONG	0.919	2026-04-09	22	1.1%	0.85	0.146
PDBC	cluster_21	Mixed: GSG, DBC, PDBC	LONG	0.918	2025-12-22	95	39.8%	0.01	0.988
DBC	cluster_21	Mixed: GSG, DBC, PDBC	LONG	0.918	2025-12-22	95	38.7%	0.02	0.984
FXF	cluster_24	Mixed: IGOV, BWX, FXE	FLAT	0.873	2026-05-05	4	0.6%	0.12	0.884
FXB	cluster_24	Mixed: IGOV, BWX, FXE	FLAT	0.873	2026-04-13	20	1.0%	0.13	0.871
IGOV	cluster_24	Mixed: IGOV, BWX, FXE	FLAT	0.873	2026-04-17	16	-0.1%	0.19	0.812
BWX	cluster_24	Mixed: IGOV, BWX, FXE	FLAT	0.873	2026-05-06	3	-0.2%	0.35	0.646
CNYA	cluster_13	Mixed: CXSE, GXC, MCHI	FLAT	0.858	2026-04-14	19	5.8%	0.00	0.996
ASHR	cluster_13	Mixed: CXSE, GXC, MCHI	FLAT	0.858	2026-04-13	20	6.9%	0.01	0.993
EWI	cluster_13	Mixed: CXSE, GXC, MCHI	FLAT	0.858	2026-04-08	23	2.3%	0.01	0.991
KBA	cluster_13	Mixed: CXSE, GXC, MCHI	FLAT	0.858	2026-04-10	21	9.8%	0.01	0.987
CNXT	cluster_13	Mixed: CXSE, GXC, MCHI	FLAT	0.858	2026-04-10	21	17.6%	0.01	0.987
FV	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-14	19	8.9%	0.00	1.000
MTUM	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	18.1%	0.00	1.000
PRN	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	15.4%	0.00	1.000
FPX	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	12.4%	0.00	1.000
RPG	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	14.2%	0.00	1.000
PDP	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	9.6%	0.00	1.000
VFMO	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	10.5%	0.00	0.998
XMMO	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-06	25	13.4%	0.00	0.998
SCHA	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	8.6%	0.00	0.998
FTC	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	9.2%	0.00	0.996
IMCG	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	9.1%	0.00	0.996
SCHM	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	7.2%	0.01	0.995
PSC	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	8.0%	0.01	0.993
IWM	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	9.2%	0.01	0.993
VTWO	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	9.2%	0.01	0.993
IWO	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	10.0%	0.01	0.992
VTWG	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	10.0%	0.01	0.992
IWC	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	10.0%	0.01	0.989
IJK	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	6.9%	0.01	0.988
MDYG	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	6.9%	0.01	0.987
IVOG	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	6.8%	0.02	0.984
VIOG	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	6.6%	0.02	0.980
QMOM	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	9.1%	0.02	0.980
SLYG	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	6.6%	0.02	0.979
IJT	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	6.6%	0.02	0.978
PAVE	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	4.4%	0.02	0.978
XSMO	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-08	23	7.6%	0.02	0.976
VBK	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	9.9%	0.03	0.975
XMHQ	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	3.6%	0.03	0.966
ISCG	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	7.4%	0.05	0.946
DWAS	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-07	24	13.7%	0.06	0.943
SMLF	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-09	22	6.4%	0.06	0.936

VXF	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	7.2%	0.07	0.932
IVOO	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	4.8%	0.07	0.927
SPMD	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	4.8%	0.07	0.926
MDY	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	4.7%	0.07	0.925
IJH	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	4.7%	0.08	0.924
VB	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	6.1%	0.09	0.914
PSCI	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	2.7%	0.10	0.896
JETS	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-05-06	3	-2.3%	0.17	0.827
VIS	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	2.1%	0.24	0.764
JHMM	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-10	21	3.9%	0.24	0.762
FIDU	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	2.1%	0.26	0.741
NUSC	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	3.0%	0.28	0.720
XLI	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	1.3%	0.34	0.664
FTXR	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-17	16	-0.5%	0.41	0.592
PKB	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	-0.2%	0.63	0.368
IYJ	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-13	20	0.7%	0.65	0.353
IYT	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-14	19	1.4%	0.82	0.185
FXR	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-04-14	19	-0.4%	0.85	0.152
XHB	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-05-06	3	-3.9%	0.98	-0.277
FXD	cluster_2	Mixed: VB, SCHM, IJH	LONG	0.849	2026-05-06	3	-3.2%	0.99	-0.388
NURE	cluster_14	Mixed: SCHH, VNQ, FREL	LONG	0.816	2026-04-13	20	5.7%	0.00	0.999
REZ	cluster_14	Mixed: SCHH, VNQ, FREL	LONG	0.816	2026-04-14	19	3.8%	0.01	0.991
USRT	cluster_14	Mixed: SCHH, VNQ, FREL	LONG	0.816	2026-04-09	22	5.2%	0.11	0.892
PSR	cluster_14	Mixed: SCHH, VNQ, FREL	LONG	0.816	2026-04-09	22	4.9%	0.12	0.878
RWR	cluster_14	Mixed: SCHH, VNQ, FREL	LONG	0.816	2026-04-10	21	4.4%	0.16	0.839
SCHH	cluster_14	Mixed: SCHH, VNQ, FREL	LONG	0.816	2026-04-09	22	4.4%	0.19	0.815
ICF	cluster_14	Mixed: SCHH, VNQ, FREL	LONG	0.816	2026-04-08	23	5.6%	0.19	0.806
FREL	cluster_14	Mixed: SCHH, VNQ, FREL	LONG	0.816	2026-04-10	21	4.0%	0.26	0.737
XLRE	cluster_14	Mixed: SCHH, VNQ, FREL	LONG	0.816	2026-04-09	22	4.3%	0.27	0.731
VNQ	cluster_14	Mixed: SCHH, VNQ, FREL	LONG	0.816	2026-04-10	21	4.0%	0.29	0.710
REET	cluster_14	Mixed: SCHH, VNQ, FREL	LONG	0.816	2026-04-13	20	3.4%	0.32	0.682
IYR	cluster_14	Mixed: SCHH, VNQ, FREL	LONG	0.816	2026-04-10	21	3.9%	0.32	0.678
RWO	cluster_14	Mixed: SCHH, VNQ, FREL	LONG	0.816	2026-04-13	20	3.0%	0.34	0.662
INDS	cluster_14	Mixed: SCHH, VNQ, FREL	LONG	0.816	2026-04-14	19	2.0%	0.51	0.489
SCHD	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-21	14	2.5%	0.00	0.999
RDIV	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-21	14	-0.2%	0.01	0.988
SPYD	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-21	14	1.0%	0.01	0.987
AMPLP	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-29	8	0.6%	0.01	0.943
RPV	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-20	15	0.3%	0.08	0.920
DIV	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-07	24	1.3%	0.12	0.879
MLPA	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-29	8	-1.5%	0.03	0.825
HDV	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-05-01	6	-1.0%	0.14	0.726
SDOG	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-07	24	1.7%	0.28	0.719
DHS	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-20	15	-0.1%	0.30	0.701
PEY	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-13	20	2.3%	0.33	0.673

SPHD	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-05-05	4	-0.5%	0.30	0.639
MLPX	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-30	7	-3.5%	0.07	0.574
ONEY	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-10	21	2.1%	0.52	0.483
DVY	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-08	23	-0.6%	0.82	0.177
CDC	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-09	22	-0.6%	0.87	0.122
EMPLP	cluster_10	Mixed: DVY, SPYD, SDOG	LONG	0.787	2026-04-28	9	-0.6%	0.56	-0.305
TDDT	cluster_19	Mixed: TDTF, TIPX, TIP	LONG	0.787	2026-04-01	27	1.2%	0.12	0.878
STPZ	cluster_19	Mixed: TDTF, TIPX, TIP	LONG	0.787	2026-04-02	26	0.9%	0.14	0.862
TDTF	cluster_19	Mixed: TDTF, TIPX, TIP	LONG	0.787	2026-04-09	22	1.0%	0.19	0.812
TIPX	cluster_19	Mixed: TDTF, TIPX, TIP	LONG	0.787	2026-04-07	24	0.8%	0.21	0.788
TIP	cluster_19	Mixed: TDTF, TIPX, TIP	LONG	0.787	2026-04-10	21	0.8%	0.22	0.777
SCHP	cluster_19	Mixed: TDTF, TIPX, TIP	LONG	0.787	2026-04-13	20	0.6%	0.23	0.769
SPIP	cluster_19	Mixed: TDTF, TIPX, TIP	LONG	0.787	2026-04-10	21	0.7%	0.27	0.732
ENZL	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-06	3	0.4%	0.01	0.993
VIDI	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-09	22	6.1%	0.02	0.984
FNDF	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-08	23	4.2%	0.02	0.976
IQDF	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	5.0%	0.03	0.973
PXF	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-08	23	4.2%	0.03	0.969
GWX	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-13	20	5.1%	0.03	0.968
VEU	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	5.0%	0.04	0.956
HFXI	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-09	22	4.7%	0.05	0.952
VXUS	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	5.1%	0.05	0.946
IXUS	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	5.1%	0.06	0.942
CWI	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	4.6%	0.06	0.939
SPDW	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	4.6%	0.07	0.933
SCHF	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	4.6%	0.07	0.930
ACWX	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	4.7%	0.07	0.927
VEA	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	4.5%	0.08	0.921
VSS	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-13	20	4.8%	0.08	0.916
FXA	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-13	20	2.1%	0.10	0.900
EWN	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	4.9%	0.11	0.891
IPKW	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-09	22	2.5%	0.13	0.868
FNDC	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	3.7%	0.13	0.867
SCZ	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	3.7%	0.16	0.844
SCHC	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-13	20	3.4%	0.20	0.803
DLS	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-13	20	2.2%	0.20	0.798
IDV	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-06	25	4.4%	0.26	0.741
FXC	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-20	15	-0.2%	0.27	0.731
FEP	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-09	22	2.7%	0.28	0.725
EWI	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-08	23	4.7%	0.28	0.716
EWV	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-05	4	3.4%	0.33	0.672
VNQI	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-04	5	2.6%	0.41	0.593
IMTM	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	2.7%	0.41	0.591
EWG	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-14	19	0.8%	0.41	0.588
EWO	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-10	21	4.2%	0.43	0.571

VIGI	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-14	19	0.5%	0.43	0.570
NFRA	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-09	22	-0.0%	0.46	0.544
VYMI	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-09	22	1.7%	0.46	0.535
BBAX	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-30	7	0.9%	0.51	0.486
DWX	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-04	5	1.0%	0.54	0.460
EWL	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-15	18	-0.2%	0.57	0.433
PID	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-30	7	-0.0%	0.58	0.414
IHDG	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-06	3	-1.5%	0.69	0.297
RWX	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-06	3	-1.4%	0.77	0.221
IDLV	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-30	7	-0.6%	0.78	0.212
EWC	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-09	22	3.0%	0.81	0.187
EZA	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-08	1	0.1%	0.67	0.182
EBND	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-06	3	0.0%	0.81	0.180
EMLC	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-06	3	-0.0%	0.84	0.151
HDEF	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-07	24	0.5%	0.93	0.014
EWD	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-05-06	3	-2.6%	0.96	-0.154
EWU	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-30	7	-1.1%	0.98	-0.348
IGF	cluster_1	Mixed: VEA, SPDW, SCHF	FLAT	0.622	2026-04-30	7	-1.6%	0.76	-0.350
SLX	cluster_12	Mixed: GNR, MXI, GUNR	FLAT	0.550	2026-04-09	22	10.8%	0.00	1.000
XME	cluster_12	Mixed: GNR, MXI, GUNR	FLAT	0.550	2026-04-15	18	7.7%	0.04	0.964
DBB	cluster_12	Mixed: GNR, MXI, GUNR	FLAT	0.550	2026-04-10	21	6.8%	0.25	0.748
URA	cluster_12	Mixed: GNR, MXI, GUNR	FLAT	0.550	2026-04-14	19	8.3%	0.35	0.647
COPX	cluster_12	Mixed: GNR, MXI, GUNR	FLAT	0.550	2026-05-08	1	1.9%	0.55	0.418
NORW	cluster_12	Mixed: GNR, MXI, GUNR	FLAT	0.550	2025-12-09	104	31.8%	0.63	0.320
NANR	cluster_12	Mixed: GNR, MXI, GUNR	FLAT	0.550	2026-05-11	0	0.0%	0.70	-0.351
IGE	cluster_12	Mixed: GNR, MXI, GUNR	FLAT	0.550	2026-05-01	6	0.1%	0.64	-0.590
MOO	cluster_12	Mixed: GNR, MXI, GUNR	FLAT	0.550	2026-05-06	3	-2.8%	0.86	-0.816
PFXF	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-14	19	4.1%	0.00	0.998
PFF	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-15	18	1.5%	0.05	0.953
HYEM	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-13	20	1.3%	0.12	0.885
VRP	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-10	21	0.5%	0.45	0.547
FPE	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-14	19	0.6%	0.46	0.536
PGX	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-17	16	-0.4%	0.58	0.421
HYLS	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-09	22	0.5%	0.64	0.363
SHYG	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-07	24	0.9%	0.76	0.243
FALN	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-10	21	0.6%	0.84	0.164
JNK	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-08	23	0.4%	0.87	0.134
SJNK	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-07	24	0.9%	0.89	0.110
HYS	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-08	23	0.1%	0.89	0.108
BBHY	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-08	23	0.3%	0.90	0.097
HYG	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-08	23	0.3%	0.91	0.090
HYLB	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-08	23	0.2%	0.92	0.077
ANGL	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-09	22	0.2%	0.95	0.053
PHB	cluster_9	Mixed: JNK, HYG, HYLB	LONG	0.489	2026-04-10	21			
IWN	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	5.8%	0.01	0.987

VTWV	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	5.7%	0.01	0.985
OMFS	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	6.3%	0.02	0.983
ISMD	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	7.5%	0.02	0.980
EES	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-08	23	5.1%	0.02	0.978
VIOO	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	5.0%	0.04	0.955
IJR	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	5.0%	0.05	0.954
SPSM	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	5.0%	0.05	0.953
PRFZ	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	6.8%	0.05	0.949
FNDA	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	5.4%	0.09	0.911
FYX	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-08	23	6.5%	0.11	0.890
FNX	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	3.9%	0.12	0.877
RWJ	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	4.6%	0.15	0.848
SLYV	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	4.5%	0.16	0.839
IJS	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	4.5%	0.17	0.831
RFV	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	2.9%	0.17	0.826
VIOV	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	4.5%	0.18	0.820
EZM	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	3.7%	0.23	0.771
VBR	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	3.3%	0.32	0.676
DES	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-08	23	3.1%	0.39	0.613
RWK	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	2.7%	0.51	0.486
RZV	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-13	20	3.5%	0.58	0.416
IJJ	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	2.2%	0.65	0.348
IVOV	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	2.1%	0.65	0.347
MDYV	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	2.2%	0.65	0.346
DON	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-13	20	0.9%	0.72	0.276
FYT	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-07	24	4.3%	0.82	0.179
XSLV	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	-0.6%	0.93	0.068
SDVY	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	1.0%	0.93	0.067
KRE	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	-1.9%	0.93	0.065
SMDV	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	-1.0%	0.94	0.061
REM	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-13	20	0.1%	0.95	0.047
KBE	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	-1.1%	0.97	0.031
REGL	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	-0.9%	1.00	0.002
IAT	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-10	21	-1.4%	1.00	0.001
SMMV	cluster_7	Mixed: VBR, MDYV, IVOV	LONG	0.380	2026-04-09	22	-1.2%	1.00	-0.056
HYD	cluster_16	Mixed: VTEB, MUB, CMF	FLAT	0.244	2026-04-08	23	0.9%	0.50	0.502
HYMB	cluster_16	Mixed: VTEB, MUB, CMF	FLAT	0.244	2026-04-09	22	0.5%	0.53	0.468
FMB	cluster_16	Mixed: VTEB, MUB, CMF	FLAT	0.244	2026-04-17	16	0.0%	0.67	0.329
NYF	cluster_16	Mixed: VTEB, MUB, CMF	FLAT	0.244	2026-04-15	18	0.4%	0.69	0.310
PZA	cluster_16	Mixed: VTEB, MUB, CMF	FLAT	0.244	2026-04-10	21	0.6%	0.73	0.273
ITM	cluster_16	Mixed: VTEB, MUB, CMF	FLAT	0.244	2026-05-08	1	0.0%	0.75	0.243
MUB	cluster_16	Mixed: VTEB, MUB, CMF	FLAT	0.244	2026-04-17	16	-0.1%	0.77	0.226
CMF	cluster_16	Mixed: VTEB, MUB, CMF	FLAT	0.244	2026-05-06	3	0.0%	0.86	0.139
TFI	cluster_16	Mixed: VTEB, MUB, CMF	FLAT	0.244	2026-05-06	3	-0.0%	0.89	0.111
SUB	cluster_16	Mixed: VTEB, MUB, CMF	FLAT	0.244	2026-05-11	0	0.0%	0.90	0.003

EWM	cluster_18	Mixed: ILF, EWZ, ECH	FLAT	0.072	2026-04-22	13	3.3%	0.00	1.000
TUR	cluster_18	Mixed: ILF, EWZ, ECH	FLAT	0.072	2026-04-10	21	3.9%	0.31	-0.694
ECH	cluster_18	Mixed: ILF, EWZ, ECH	FLAT	0.072	2026-05-06	3	-2.0%	0.98	-0.170
RSX	cluster_18	Mixed: ILF, EWZ, ECH	FLAT	0.072	2023-07-27	699			
PSCH	cluster_11	Mixed: VHT, FHLC, FXH	LONG	0.021	2026-04-13	20	4.8%	0.06	0.943
IBB	cluster_11	Mixed: VHT, FHLC, FXH	LONG	0.021	2026-05-06	3	-2.2%	0.81	-0.006
LDUR	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-04-07	24	0.4%	0.29	0.708
CMBS	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.1%	0.41	0.575
IAGG	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.3%	0.42	0.533
BNDX	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.3%	0.41	0.532
SPSB	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-04-09	22	0.1%	0.71	0.287
IGLB	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.4%	0.76	0.196
USIG	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-06	3	-0.2%	0.81	0.175
VCLT	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.4%	0.78	0.167
GTO	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-07	2	0.0%	0.80	0.166
FLCO	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-07	2	0.3%	0.81	0.162
SPLB	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.3%	0.78	0.159
CORP	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-06	3	-0.2%	0.83	0.153
GNMA	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-07	2	-0.1%	0.79	0.111
BAB	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.65	0.098
IGIB	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-07	2	0.1%	0.87	0.087
SPIB	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-06	3	-0.1%	0.90	0.081
LQD	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.2%	0.85	0.067
AGGY	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.3%	0.84	0.038
VCIT	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-07	2	0.1%	0.90	0.034
IUSB	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.2%	0.87	0.020
BSV	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-06	3	-0.1%	0.91	0.019
QLTA	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.2%	0.90	0.002
FBND	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-08	1	-0.3%	0.87	-0.047
VMBS	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.86	-0.083
BOND	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.91	-0.105
MBB	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.87	-0.169
GVI	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.93	-0.224
AGG	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.89	-0.232
BND	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.90	-0.256
SPAB	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.89	-0.299
SCHZ	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.93	-0.343
SPMB	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.89	-0.358
AGZ	cluster_5	Mixed: BND, AGG, SCHZ	LONG	-0.177	2026-05-11	0	0.0%	0.98	-0.526
SIVR	cluster_15	Mixed: DBP, GLTR, GDXJ	FLAT	-0.193	2026-05-11	0	0.0%	0.15	0.796
SLV	cluster_15	Mixed: DBP, GLTR, GDXJ	FLAT	-0.193	2026-05-11	0	0.0%	0.15	0.794
PPLT	cluster_15	Mixed: DBP, GLTR, GDXJ	FLAT	-0.193	2026-05-11	0	0.0%	0.50	0.155
SILJ	cluster_15	Mixed: DBP, GLTR, GDXJ	FLAT	-0.193	2026-05-11	0	0.0%	0.52	0.141
SIL	cluster_15	Mixed: DBP, GLTR, GDXJ	FLAT	-0.193	2026-05-11	0	0.0%	0.59	0.096
XLP	cluster_17	Mixed: VDC, FSTA, XLP	LONG	-0.232	2026-05-05	4	-0.8%	0.00	0.999

KXI	cluster_17	Mixed: VDC, FSTA, XLP	LONG	-0.232	2026-05-06	3	-1.6%	0.02	0.980
IYK	cluster_17	Mixed: VDC, FSTA, XLP	LONG	-0.232	2026-05-06	3	-0.2%	0.01	0.974
VDC	cluster_17	Mixed: VDC, FSTA, XLP	LONG	-0.232	2026-05-01	6	-1.7%	0.04	0.960
FSTA	cluster_17	Mixed: VDC, FSTA, XLP	LONG	-0.232	2026-05-01	6	-1.8%	0.04	0.956
PSCC	cluster_17	Mixed: VDC, FSTA, XLP	LONG	-0.232	2026-04-28	9	-2.2%	0.77	0.223
UUP	cluster_26	Mixed: UUP, SQQQ	FLAT	-1.000	2026-04-27	10	-0.4%	0.47	-0.466

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	composite_rank
RSX			cluster_18	11	LONG	0.371	0.052	0.00021	2.5
UNG			cluster_18	11	FLAT	0.492	0.081	0.00017	3.5
GBIL			cluster_23	9	FLAT	0.692	0.040	0.00007	7.0
VNLA			cluster_23	9	LONG	0.635	0.221	0.00034	11.5
FXV			cluster_24	6	FLAT	0.732	0.094	0.00022	11.5
DBA			cluster_18	11	FLAT	0.591	0.174	0.00038	15.8
FTSM			cluster_23	9	FLAT	0.787	0.130	0.00030	18.2
BIL			cluster_23	9	FLAT	0.825	0.021	0.00000	20.8
SHV			cluster_23	9	FLAT	0.825	0.089	0.00022	23.2
NEAR			cluster_23	9	FLAT	0.720	0.182	0.00041	23.5
MINT			cluster_23	9	FLAT	0.656	0.188	0.00047	24.8
ICSH			cluster_23	9	FLAT	0.793	0.162	0.00038	26.2
TUR			cluster_18	11	LONG	0.487	0.219	0.00056	31.2
GSY			cluster_23	9	LONG	0.793	0.179	0.00043	31.5
LDUR			cluster_5	58	LONG	0.844	0.124	0.00033	32.0
FXF			cluster_24	6	LONG	0.818	0.203	0.00047	38.2
CMBS			cluster_5	58	LONG	0.876	0.160	0.00030	39.5
LMBS			cluster_5	58	FLAT	0.836	0.231	0.00044	41.2
SHM			cluster_16	13	FLAT	0.877	0.250	0.00044	52.8
VNM			cluster_18	11	FLAT	0.452	0.334	0.00082	55.0
PPLT			cluster_15	13	LONG	0.750	0.269	0.00072	56.5
SUB			cluster_16	13	LONG	0.877	0.263	0.00051	57.0
KURE			cluster_13	15	FLAT	0.804	0.271	0.00070	61.2
DBB			cluster_12	17	LONG	0.751	0.357	0.00082	64.5
EIDO			cluster_18	11	FLAT	0.514	0.374	0.00105	71.5
EPHE			cluster_18	11	FLAT	0.578	0.409	0.00105	72.0
KBA			cluster_13	15	LONG	0.855	0.254	0.00070	72.8
FLTR			cluster_22	9	FLAT	0.864	0.332	0.00070	75.8
ECH			cluster_18	11	LONG	0.637	0.429	0.00116	81.2
CNXT			cluster_13	15	LONG	0.905	0.242	0.00062	81.5

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20